

# Planned Open Source tools and Libraries

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## Orchestration

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- **LangChain** — Core agent orchestration framework providing tool chaining, prompt templating, document loaders, and memory management.
- **LangGraph** — Stateful multi-step agentic workflow graph that controls the ReAct agent loop and tool-use decision branching.

## LLM / Models

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- **Mistral 7B / LLaMA 3 / Gemma 4** — Self-hosted open-source LLMs for answer generation, ensuring full data sovereignty with no external API dependency.
- **HuggingFace Transformers** — Handles model loading, tokenization, and inference pipelines for local LLM and embedding models.
- **sentence-transformers** — Generates dense embeddings for documents and queries locally using models such as all-MiniLM-L6-v2 and BGE-M3.

## Vector Database

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- **Qdrant** — Primary vector database for storing embeddings with metadata, supporting hybrid search, sharding, and replication.
- **ChromaDB** — Lightweight alternative vector store suited for local development and rapid prototyping.
- **Weaviate** — In-memory vector search option optimised for smaller corpora and fast retrieval benchmarking.

## Retrieval

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- **rank\_bm25** — Provides BM25 keyword search for hybrid retrieval, combined with vector search for improved recall.
- **flashrank / cross-encoder** — Reranks retrieved document chunks by relevance before they are passed into the LLM context window.

## Document Processing

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- **MinIO** — High-performance, S3-compatible open-source object storage for storing large volumes of unstructured data such as documents, embeddings, and model artifacts on-premises.
- **PyMuPDF (fitz)** — Extracts text and metadata from PDFs, handling both scanned and structured files with page-level tracking.
- **pdfplumber** — Specialised in table extraction from PDFs, used in the IDP Classifier for document structure detection.
- **python-docx** — Reads and parses .docx Word documents for use in the ingestion pipeline.
- **unstructured** — Unified document ingestion library that handles PDF, DOCX, HTML, CSV, and TXT in a single pipeline.
- **Qwen 3.5 Vision / LLaMA 4 Maverick** — Self-hosted vision language models for image processing, maintaining full data sovereignty with no external API dependency.

- **tiktoken** — OpenAI's tokenizer used for accurate token-count-aware chunking of documents before ingestion.

## API & Backend

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- **FastAPI** — High-performance async REST API framework that serves all agent endpoints and webhooks.
- **Uvicorn** — ASGI production server that runs FastAPI with high concurrency.
- **Pydantic** — Enforces data validation and schema definitions across all API request and response models.
- **SQLAlchemy** — ORM layer for PostgreSQL, managing ticket metadata, user sessions, and feedback logs.
- **Redis** — In-memory cache for session and conversation memory as well as rate limiting.

## Frontend / UI

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- **Streamlit** — Rapid chat UI prototype providing the user-facing query interface for demos.
- **React.js** — Production-grade web frontend powering the Admin Dashboard and Chat UI.

## Storage & Database

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- **PostgreSQL** — Relational database storing ticket metadata, user profiles, RBAC configurations, logs, and feedback.

## Evaluation

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- **RAGAS** — RAG evaluation framework measuring faithfulness, answer relevancy, and context precision and recall.
- **deepeval** — LLM evaluation library for custom metrics and A/B comparisons across retrieval strategies.
- **scikit-learn** — Computes F1 score, precision, and recall for quantitative retrieval quality metrics.

## DevOps & Infrastructure

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- **Docker** — Containerises all services — LLM, vector database, API, and UI — for consistent and portable deployment.
- **Kubernetes (K8s)** — Orchestrates containers at scale with horizontal pod autoscaling across all services.
- **Prometheus** — Collects system metrics including query latency, retrieval hit rate, and ticket deflection rates.
- **Grafana** — Visualises Prometheus metrics through dashboards for system health and performance monitoring.
- **GitHub Actions** — CI/CD pipeline automating testing, Docker image builds, and deployment on each code push.